

## Excercises

1. Find mean, median, mode and range for: 12, 15, 18, 15, 20, 22, 15, 25.
2. Construct a frequency distribution for: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4.
3. Find Q1, Q2, Q3 and IQR for: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30.
4. Detect outliers using the IQR method for: 10, 12, 13, 15, 16, 17, 18, 20, 45.
5. Determine the skewness (Asymmetry) of: 5, 6, 7, 7, 8, 8, 9, 20.
6. A uniformly distributed variable ranges from 10 to 50. Find  $P(20 < X < 35)$ .
7. A normal distribution has mean 100 and standard deviation 15. Find the Z-score for  $X=130$ .
8. For the same normal distribution, find  $P(X < 130)$ .
9. Given a dataset with one extreme value, compare the mean and median before and after outlier treatment.
10. Perform a complete EDA on a small student-mark dataset and report central tendency, dispersion, distribution, outliers and skewness.

## Solutions

### 1. Mean, Median, Mode and Range

Given data

12, 15, 18, 15, 20, 22, 15, 25

#### Step 1: Mean

Formula:

$$\text{Mean} = \sum X / n$$

$$\sum X = 12 + 15 + 18 + 15 + 20 + 22 + 15 + 25 = 142$$

Number of values:

$$n=8$$

Therefore,

$$\text{Mean} = 142 / 8$$

$$=17.75$$

### **Step 2: Median**

Arrange the data:

12,15,15,15,18,20,22,25

There are 8 values, so:

$$\text{Median} = \frac{4^{\text{th}} + 5^{\text{th}}}{2}$$

$$= \frac{15 + 18}{2}$$

$$=16.5$$

### **Step 3: Mode**

The value occurring most frequently is **15**.

$$\text{Mode} = 15$$

### **Step 4: Range**

$$\text{Range} = \text{Maximum} - \text{Minimum} = 25 - 12 = 13$$

**Answer**

| Measure | Value        |
|---------|--------------|
| Mean    | <b>17.75</b> |
| Median  | <b>16.5</b>  |

|       |           |
|-------|-----------|
| Mode  | <b>15</b> |
| Range | <b>13</b> |

## 2. Frequency Distribution

### Given data

2,3,4,2,5,3,4,2,6,5,3,4

A **frequency distribution** shows how many times each value occurs.

Count each value:

| Value | Frequency |
|-------|-----------|
| 2     | 3         |
| 3     | 3         |
| 4     | 3         |
| 5     | 2         |
| 6     | 1         |

Check:

$$3+3+3+2+1=12$$

So the total frequency is **12**.

### Answer

The most frequent values are **2, 3 and 4**, each occurring **3 times**.

### **3. Q1, Q2, Q3 and IQR**

#### **Given data**

5,8,10,12,15,18,20,22,25,30

The data is already sorted.

There are 10 values.

#### **Q2 — Median**

$$Q2 = (15 + 18) / 2$$

$$Q2 = 16.5$$

#### **Q1**

Lower half:

5,8,10,12,15

Middle value:

$$Q1 = 10$$

#### **Q3**

Upper half:

18,20,22,25,30

Middle value:

$$Q3 = 22$$

#### **IQR**

Formula:

$$IQR = Q3 - Q1$$

$$IQR = 22 - 10$$

$$IQR = 12$$

**Answer**

$Q1 = 10$  ,  $Q2 = 16.5$  ,  $Q3 = 22$  ,  $IQR = 12$

**4. Detect Outliers Using IQR****Given data**

10,12,13,15,16,17,18,20,45

**Step 1: Find Q1**

Lower half:

10,12,13,15

$$Q1 = (12 + 13) / 2$$

$$Q1 = 12.5$$

**Step 2: Find Q3**

Upper half:

17,18,20,45

$$Q3 = (18 + 20) / 2$$

$$Q3 = 19$$

**Step 3: Calculate IQR**

$$IQR = Q3 - Q1$$

$$= 19 - 12.5$$

$$= 6.5$$

**Step 4: Find upper boundary**

$$\text{Upper Bound} = Q3 + 1.5(IQR)$$

$$= 19 + 1.5(6.5)$$

$$=19+9.75$$

$$\text{Upper bound}=28.75$$

Any value greater than **28.75** is an outlier.

The value:

$$45 > 28.75$$

Therefore,

45 is an outlier

**Answer**

**Outlier = 45**

## **5. Determine Skewness**

**Given data**

5,6,7,7,8,8,9,20

The value **20** is much larger than the other observations.

Mean:

$$\text{Mean} = 70 / 8 = 8.75$$

Median:

$$\text{Median} = (7+8)/2 = 7.5$$

Since:

$$\text{Mean} > \text{Median}$$

the distribution is **positively skewed (right-skewed)**.

Using Pearson's coefficient:

$$\text{Skewness} = \text{SD}^3(\text{Mean} - \text{Median}) = 4.413(8.75 - 7.5) \approx 0.85$$

So the data is:

Positively / right skewed

**Answer**

The value **20** pulls the distribution toward the right, making it **right-skewed**.

## **6. Uniform Distribution**

**Given**

A uniformly distributed variable ranges from:

$$10 \leq X \leq 50$$

Find:

$$P(20 < X < 35)$$

For a uniform distribution:

$$P(a < X < b) = \frac{b - a}{\text{Maximum} - \text{Minimum}}$$

Therefore:

$$P(20 < X < 35) = \frac{35 - 20}{50 - 10} = \frac{15}{40} = 0.375$$

As a percentage:

$$0.375 \times 100 = 37.5\%$$

**Answer**

$$P(20 < X < 35) = 0.375 = 37.5\%$$

## **7. Find Z-Score**

**Given**

Mean:

$$\mu=100$$

Standard deviation:

$$\sigma=15$$

Value:

$$X=130$$

**Formula**

$$Z = \frac{X - \mu}{\sigma}$$

Substitute:

$$Z = \frac{130 - 100}{15}$$

$$Z = 30/15$$

$$Z = 2$$

**Answer**

The value **130** is **2 standard deviations above the mean**.

**8. Find  $P(X < 130)$**

From Question 7:

$$Z = 2$$

We need:

$$P(X < 130) = P(Z < 2)$$

From the standard normal table:

$$P(Z < 2) = 0.9772$$

Therefore:

$$P(X < 130) = 0.9772$$



Percentage:

$$0.9772 \times 100 = 97.72\%$$

**Answer**

Approximately **97.72% of observations are below 130.**

## **9. Effect of an Outlier on Mean and Median**

Consider this example:

**Original data**

10,12,14,15,16

Mean:

$$510 + 12 + 14 + 15 + 16 = 567 = 13.4$$

Median:

14

Now suppose **100** is an extreme outlier.

**Data with outlier**

10,12,14,15,16,100

Mean:

$$10 + 12 + 14 + 15 + 16 + 100 / 6 = 167 / 6 = 27.83$$

Median:

$$(14 + 15) / 2 = 14.5$$

**Comparison**

| Measure | Before outlier | After outlier |
|---------|----------------|---------------|
|---------|----------------|---------------|

|        |      |              |
|--------|------|--------------|
| Mean   | 13.4 | <b>27.83</b> |
| Median | 14   | <b>14.5</b>  |

## Conclusion

The **mean changes dramatically** because it is sensitive to extreme values.

The **median changes only slightly**.

Therefore:

Median is more resistant to outliers than mean

## 10. Complete EDA on Student Marks

For demonstration, consider this student-mark dataset:

45,50,52,55,58,60,62,65,68,95

EDA means **Exploratory Data Analysis**. It is the process of examining data to understand its characteristics, distribution, variability and unusual observations.

### A. Central Tendency

#### Mean

$$\text{Mean} = (45 + 50 + 52 + 55 + 58 + 60 + 62 + 65 + 68 + 95) / 10$$

$$\text{Mean} = 61$$

#### Median

There are 10 observations:

$$\text{Median} = (58 + 60) / 2$$

$$\text{Median} = 59$$

## **Mode**

Every value occurs only once.

No mode

## **B. Dispersion**

### **Range**

Range=95–45

Range=50

Using the **sample** formula, the variance is approximately:

Variance=191.78

Standard deviation:

SD=13.85

This indicates that the marks vary by roughly 14 marks around the mean.

## **C. Quartiles**

Using the usual percentile calculation:

Q1=52.75

Q2=59

Q3=64.25

Therefore:

IQR=Q3–Q1

=64.25–52.75

IQR=11.5

#### **D. Outlier Detection**

Upper boundary:

$$Q3 + 1.5(IQR)$$

$$= 64.25 + 1.5(11.5)$$

$$= 64.25 + 17.25$$

$$= 81.5$$

Any mark above **81.5** is an outlier.

Since:

$$95 > 81.5$$

95 is an outlier

#### **E. Skewness**

The value **95** is much higher than the other marks.

Also:

$$\text{Mean} = 61 > \text{Median} = 59$$

Therefore, the distribution is:

Positively / right skewed

The calculated skewness is approximately:

$$1.47$$

which indicates noticeable right skewness.